

ENGINEERING SPECIFICATIONS

Document Name: VFD Set Points- KAS/KWS/KWE/KWK With Danfoss make VFD(FC102) Date: 20/10/2023

Document No: ESP-20-21-002 Revision: R01 (MODBUS PARAMETERS UPDATED)

Prepared By: ASP Revised By: IGP Approved By: SSN Sheet: 1 of 2

ID	Motor Parameters	Value
0 30	Custom Readout Unit	RPM
0 31	Custom Readout Min Value	0
0 32	Custom Readout Max Value	3000
0 60	Main Menu password	2020
0 61	Access to main menu	LCP read only
0 40	[Hand on] Key on LCP	Disabled
1 03	Torque Characteristics	Compressor torque
1 20	Motor power (KW)	**
1 22	Motor Voltage	**
1 23	Motor Frequency	**
1 24	Motor Current	**
1 25	Motor Nominal Speed	**
3 03	Maximum Reference	51
3 13	Reference Site	Remote
3 15	Reference 1 Source	Analog input 53
3 16	Reference 2 Source	No function
3 17	Reference 3 Source	No function
3 41	Ramp 1 Ramp Up Time	8
3 42	Ramp 1 Ramp Down Time	10
4 10	Motor Speed Direction	Clockwise
4 12	Motor Speed Low Limit [Hz]	22
4 14	Motor Speed High Limit [Hz]	51
4 16	Torque Limit Motor Mode	120
4 18	Current Limit	$\left[\frac{\text{RLA of Comp (as per SO)}}{\text{Motor Current (para. 1 24)}} \times 100 \right] + 10$
4 19	Max Output Frequency	51
4 50	Warning Current Low	0
4 51	Warning Current High	Showing Auto
4 52	Warning Speed Low	0
4 53	Warning Speed High	**
5 13	Terminal 29 Digital Input	No operation
5 40	Function Relay	Drive ready
5 40 1	Function Relay	Running
6 15	Terminal 53 High Ref./ Feedback Value	51
6 50	Terminal 42 Output	Motor cur. 0-lmax

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8 01	Control Site	Digital and ctrl.word
8 02	Control Source	FC Port
8 10	Control Profile	FC profile
8 30	Protocol	Modbus RTU
8 31	Address	1,2 (As per compressor/Circuit number)
8 32	Baud Rate	19200 Baud
8 33	Parity / Stop Bits	No parity , 1 Stop Bit (For Carel Controller)
		Even parity , 1 Stop Bit (For Siemens Controller)
8 43 0	PCD Read Configuration	Status Word [1603]
8 43 1	PCD Read Configuration	Main Actual Value [%] [1605]
8 43 2	PCD Read Configuration	Reference [%] [1602]
8 43 3	PCD Read Configuration	Power [kW] [1610]
8 43 4	PCD Read Configuration	Motor voltage [1612]
8 43 5	PCD Read Configuration	Frequency [1613]
8 43 6	PCD Read Configuration	Motor current [1614]
8 43 7	PCD Read Configuration	DC Link Voltage [1630]
8 43 8	PCD Read Configuration	Heatsink Temp. [1634]
14 25	Trip Delay at Torque Limit	25
14 26	Trip Delay at Inverter Fault	2
16 37	Inverter max current	Showing Auto
22 43	Wake-up Speed [Hz]	30

NOTES

- 1) FLA SETPOINT IN CONTROLLER - $RLA(AS\ PER\ SO/CHILER\ NAMEPLATE) \times 0.87$
 ** 2) REFER ESP-20-21-022_MOTOR DATA FOR VFD SETPOINT